

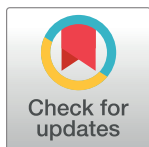
RESEARCH ARTICLE

Prognostic value of procalcitonin in acute exacerbation of chronic obstructive pulmonary disease: A systematic review and meta-analysis

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Abstract

Aims

To evaluate the prognostic role of procalcitonin(PCT) on all-cause mortality in acute exacerbation of chronic obstructive pulmonary disease (AECOPD).

Methods

Database including PubMed, EMBASE, and the Cochrane Central Register of Controlled Trials were searched to find relevant trials. We included studies with patients hospitalized for AECOPD, which assessed procalcitonin levels and reported on the association between procalcitonin and mortality..

Results

Fourteen trials involving 2983 patients were included. We found PCT levels in patients hospitalized for AECOPD are not associated with increased risk of mortality (RR 1.03, 95%CI 0.99–1.08). However, subgroup analysis showed PCT levels are indeed associated with an increased risk of mortality in mild to moderate AECOPD(RR 1.74, 95%CI 1.11–2.73). Deceased patients had significantly higher PCT levels, compared to survivors (MD 0.61, 95%CI 0.30–0.92). In PCT positive group, there was a significant increase in all-cause mortality(OR 3.21, 95%CI 1.84–5.61).

Conclusions

Results from this meta-analysis suggest that procalcitonin levels at the time of hospital admission for mild to moderate AECOPD are positively correlated with mortality.

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Competing interests: The authors have declared that no competing interests exist.

Introduction

Chronic obstructive pulmonary disease (COPD) is a common, preventable, and treatable disease that characterized by persistent respiratory symptoms and irreversible airflow limitation [1, 2]. Globally, COPD is the third most common cause of mortality, contributing to 6% of total deaths in 2019, with a majority occurring in low- and middle-income nations [3, 4]. Acute exacerbations of chronic obstructive pulmonary disease (AECOPD) play a significant role in the management of COPD, due to its negative impacts on health status and disease progression [5]. Hospitalization for COPD exacerbation is linked to an unfavorable prognosis and an elevated mortality risk [6].

Numerous investigations have been conducted to ascertain the prognostic indicators of mortality in patients with acute exacerbations of chronic obstructive pulmonary disease (AECOPD). It appears that that several clinical factors are associated with mortality from AECOPD, such as age, lower body mass index(BMI), forced expiratory volume in one second (FEV1), cardiac failure, diabetes mellitus, ischemic heart disease, malignancy, long-term oxygen requirement, and partial pressure of oxygen (PaO₂) upon admission [7, 8]. However, there is little information about the relationship between biomarkers and death in COPD. Given above, effective biomarkers are sought for determining the acute attack frequency, length of hospitalization, severity of morbidity, and mortality.

Procalcitonin(PCT) is a small protein that is normally undetectable in plasma, which is significantly increased in the bacterial infections [9]. It has been shown that PCT-guided antibiotic therapy can safely reduce antibiotic overuse in patients with AECOPD [10]. PCT has been reported to be related to mortality in pneumonia and severe sepsis [11, 12]. Therefore, we conducted a comprehensive systematic review and meta-analysis to assess the value of procalcitonin in predict mortality among patients hospitalized for AECOPD.

Methods

The present study was performed in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-analyses(PRISMA) statement (S1 Checklist) and the Meta-analysis of Observational Studies in Epidemiology(MOOSE) checklist (S1 File) [13, 14]. The protocol for this study has been published on PROSPERO (registration number CRD42020158430).

Search strategy and selection criteria

We conducted a systematic search of the PubMed, EMBASE and Cochrane Library databases for articles published up to October 2023 in any language. The search strategies were as follows: (COPD or “chronic obstructive pulmonary disease” or “chronic airflow obstruction” or “chronic obstructive airway disease” or “chronic obstructive lung disease”) and (procalcitonin or PCT or pro-calcitonin or “calcitonin precursor polypeptide” or “calcitonin 1” or “calcitonin related polypeptide alpha”) (S1 Table). Additional data sources were examined, including conference proceedings and the reference lists of the relevant studies.

Studies fulfilling the following selection criteria were included in this meta-analysis: (1) the type of study design was observational research, (2) the relationship between procalcitonin and mortality in patients with AECOPD, (3) diagnosis of COPD based on guidelines from the Global Initiative for Chronic Obstructive Lung Disease (GOLD). Reviews, case reports, conference abstracts, and animal experiments were excluded. If multiple studies used the same patient sample, the most recent or informative article was included. Two reviewers (Lin C and Li G) independently performed the search strategy and evaluated the studies. Any disagreement was resolved by a third reviewer (Pang Q).

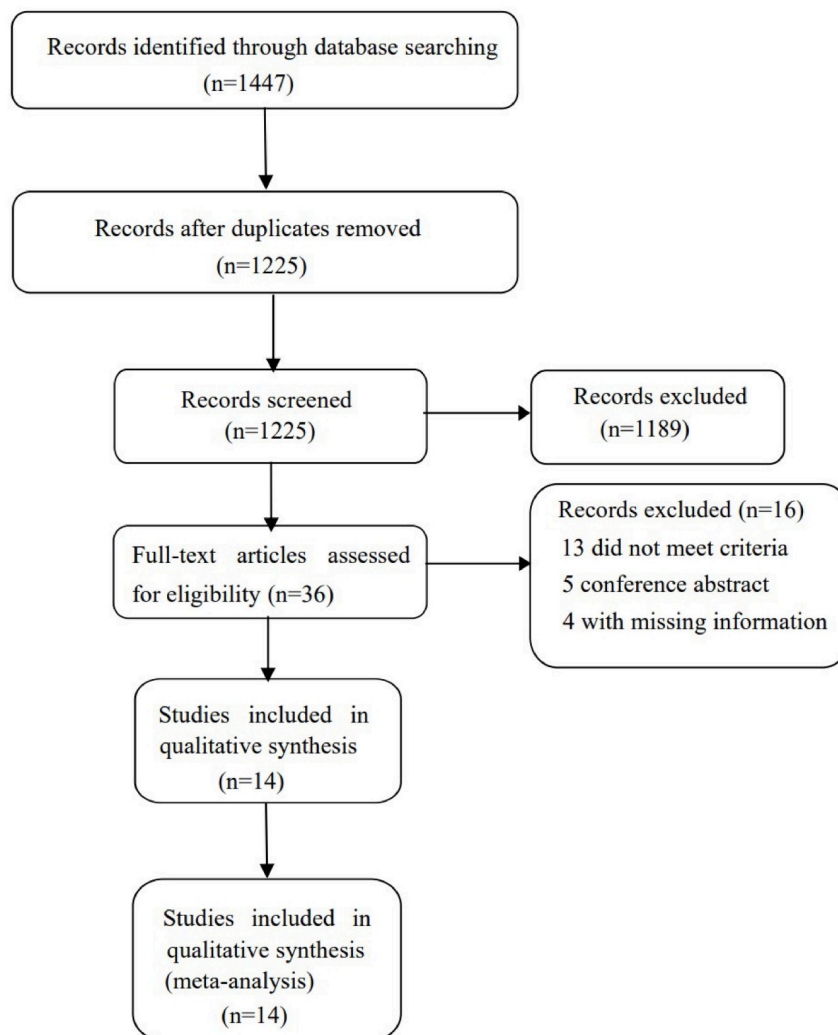


Fig 1. Diagram illustrating the process for identifying relevant trials.

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Study selection

The process of identifying relevant trials is shown in Fig 1. A search of databases revealed a total of 1447 studies, after removal of duplicates there were 1225 studies left for analysis. During the screening process, 1189 studies were eliminated for various reasons based on the title and abstract. Subsequently, fourteen studies were included in our analysis [15–26], and two [15, 23] of which used the same sample. All studies identified after excluding duplications and reasons for exclusion of each study is shown in S2 Table. Eight studies reported the ability of PCT predicts mortality in AECOPD. Five studies compared mortality according to PCT levels. And four studies compared PCT levels between survivors and deceased. Eleven studies were conducted in Europe, and three in Asia. The characteristics of the included trials are showed in Table 1. All data extracted in primary studies is shown in S3 Table.

Date extraction

Two authors (Lin C and Li G) independently reviewed the full manuscripts of eligible trails, and the relevant data were extracted into predesigned data collection forms. We verified the

Table 1. Characteristic of included trials associating with mortality.

Author	Year	Clinical Setting	Study design	Sample	Follow-up
Stolz	2007	ED	Prospective Study	167	Hospital/6 months
	2008				2 years
Rammaert	2009	ICU	Prospective Study	116	In-hospital
Zuur-Telgen	2014	Hospitalization	Prospective Study	181	3 years
Ceylan	2015	Hospitalization	Not Mentioned	58	In-hospital
Grolinund	2015	Hospitalization	Prospective Study	217	5–7 years
Kutz	2015	Hospitalization	Prospective Study	584	30 days
Ergan	2016	ICU	Retrospective Study	63	In-hospital
Flattet	2017	Hospitalization	Retrospective Study	359	5 years
Gong	2020	Hospitalization	Prospective Study	110	6 months
Yu	2020	Hospitalization	Prospective Study	695	In-hospital
Galani	2021	ICU	Retrospective Study	127	28 days
Yao	2021	Hospitalization	Retrospective Study	146	28 days
Koc	2022	Hospitalization	Prospective Study	160	6 months

ED: emergency department

ICU: intensive care unit

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accuracy of relevant data by comparing the collection forms. Any discrepancies between two reviewers were resolved through the evaluation of a third reviewer (Pang Q). The following data were collected from each study: author, year, study design, location, sample size, follow-up, all factors considered at multivariate analysis. Authors of included studies were contacted via email if further study details were needed.

Qualitative assessment

To assess the quality of the eligible studies, two independent authors (Lin C and Li G) scored the studies according to the Newcastle-Ottawa Scale (NOS) for cohort studies. The 9-point NOS contains three items: selection (0–4), comparability (0–2), and exposure/outcome (0–3) (S4 Table). Studies that scored over 7 points on the NOS were deemed to be of high quality. Any disagreement was resolved via the evaluation of a third reviewer (Pang Q).

Statistical analysis

For PCT predicts mortality, point estimates and standard errors were extracted from individual studies and were combined using the generic inverse variance method. The estimates of log ORs and standard errors have been obtained from the results of Cox proportional hazards regression models of the included studies. Subgroup analysis was performed based on the severity of disease. Severe cases are defined as requiring intubation or mechanical ventilation, and others as mild to moderate.

Continuous variables were pooled as mean difference (MD) and 95% confidence interval (CI) using inverse variance method. Mortality was analyzed using the Mantel-Haenszel (M-H) method to calculate odds ratio (OR) and 95% confidence interval (CI). The outcome measure was assessed in an intention-to-treat (ITT) manner. P-values < 0.05 were considered statically significant. Considering the high likelihood of between-study variance, we used a random effect model. Heterogeneity among studies was assessed by Cochran Q and I^2 statistics, I^2

values of more than 50% were considered to represent significant heterogeneity [27]. All data analyses were performed using the RevMan(v 5.3) software.

Results

PCT predict mortality

Eight clinical trials [15–22] with a total of 2085 patients reported that the level of PCT predicts mortality in patients with AECOPD. Five of these trials were prospective and three were retrospective. All trials showed that an elevated PCT level was associated with a higher risk of all-cause mortality, with hazard ratios (HR) ranging from 1.00 to 6.12. The pooled RR was 1.03 (95% CI, 0.99–1.08) (Fig 2). There was high heterogeneity across studies for this outcome, with $I^2 = 79\%$ (95% CI, 58%–89%).

Of these, five of the trials evaluated patients admitted to hospital or emergency department due to AECOPD, and the other three trials evaluated severe case of AECOPD requiring intubation and mechanical ventilation. We performed subgroup analysis based on the severity of AECOPD (Fig 3). The pooled RR for severe AECOPD was 1.01 (95% CI, 0.97–1.05), with moderate heterogeneity ($I^2 = 65\%$; 95% CI 0%–90%). The pooled RR for mild to moderate AECOPD was 1.74 (95% CI, 1.11–2.73), with high heterogeneity ($I^2 = 85\%$; 95% CI 67%–93%).

Mortality in different levels of PCT

Five trials with 514 patients compared mortality between groups according to PCT levels. The PCT cutoff levels was 0.25ng/ml for four of the trials, and 0.10ng/ml for the remaining trial. There was a significant increase in all-cause mortality in PCT positive group (OR 3.21, 95% CI 1.84–5.61; $I^2 = 0\%$) (Fig 4).

Comparison of PCT levels between deceased and survivals

Four trials with 1566 patients compared PCT levels between survivors and deceased. The pooled data showed that deceased patients had significantly higher levels of PCT, compared to survivors (MD 0.61, 95% CI 0.30–0.92) (Fig 5). There was moderate heterogeneity across studies, with $I^2 = 52\%$ (95% CI, 0%–84%).

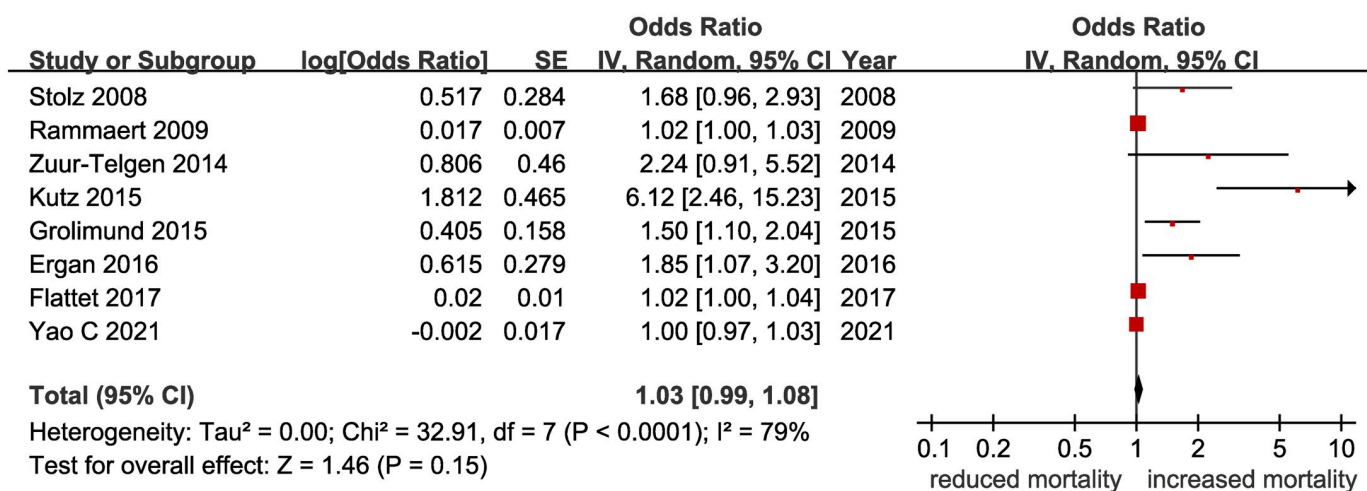


Fig 2. The association between procalcitonin levels and mortality in patients with AECOPD.

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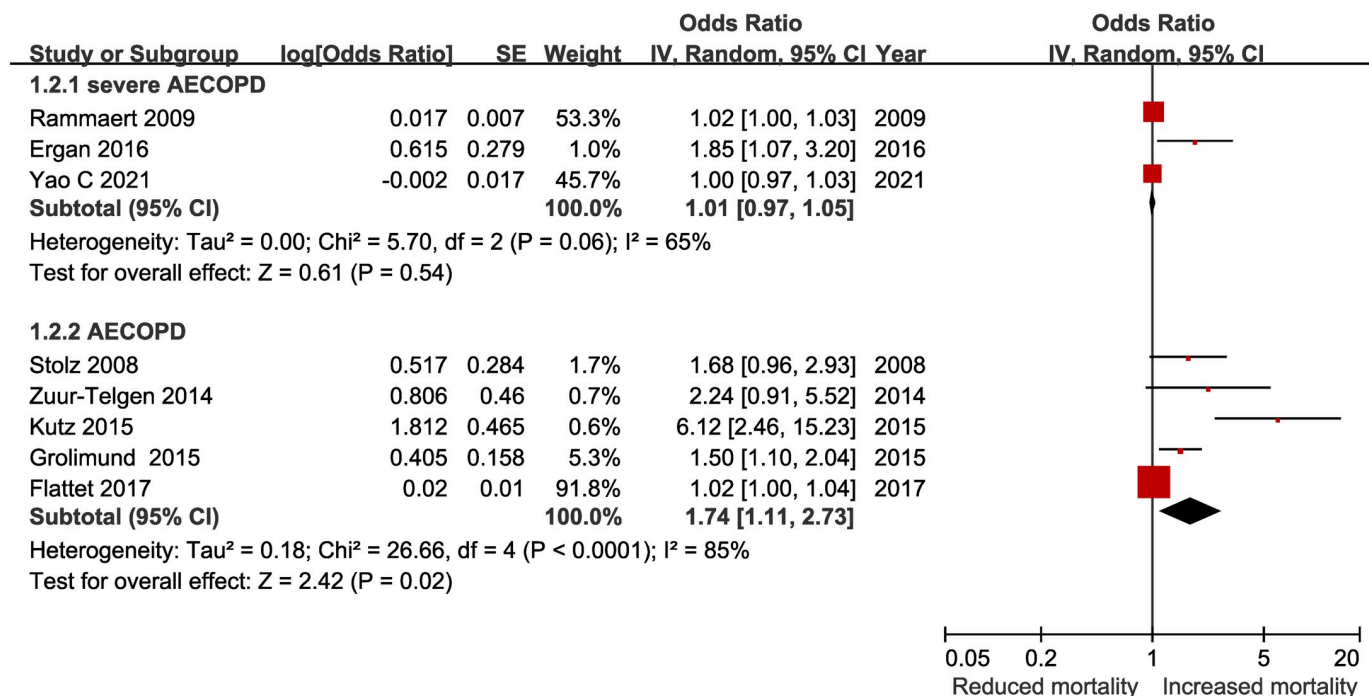


Fig 3. The association between procalcitonin levels and mortality in patients with AECOPD stratified by the severity of disease.

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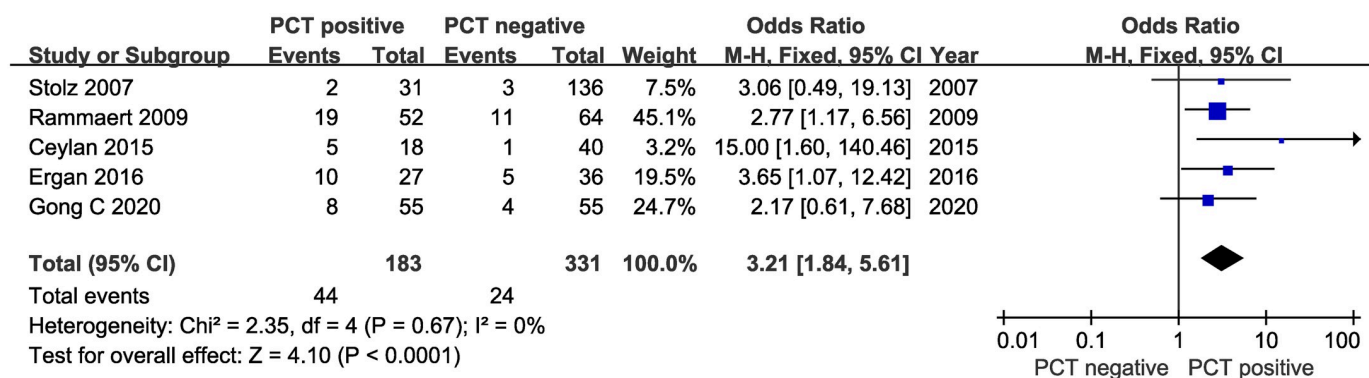


Fig 4. Meta-analysis of studies compared mortality between groups according to PCT levels.

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Discussion

In the current meta-analysis, the results determined that PCT levels in patients hospitalized for AECOPD are not associated with increased risk of mortality. However, we found that mortality is significantly higher in PCT positive group ($P < 0.05$), and the PCT levels in deceased patients are significantly higher than those in survivors ($P < 0.05$).

COPD is a prevalent chronic disease, with high mortality and morbidity. The frequency and severity of acute exacerbation are the most important factors determining overall prognosis in COPD, which both short- and long-term mortality rates are increased [28]. Therefore, early identification of high-risk patients for mortality is particularly important for improving

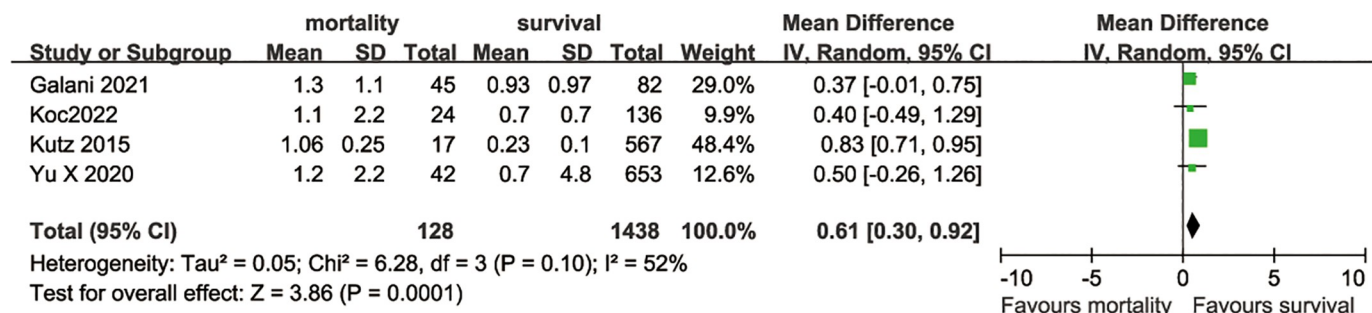


Fig 5. Meta-analysis of studies compared PCT levels between survivors and deaths.

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prognosis. However, there is no single inflammatory marker used to predict the prognosis of AECOPD. Bacterial infection is a leading reason for AECOPD, and antibiotic treatment is necessary for the patients [29]. PCT is a specific marker of systemic bacterial infection and its levels correlate with the severity in critically ill patients [18, 30]. Several studies have suggested that the level of PCT in patients with AECOPD are higher than those in stable stage [31, 32]. Wang et al. [33] showed PCT levels are positively correlated with the severity of COPD. It is reasonable to assume that the level of PCT is associated with an increased risk of death. In our review, PCT levels are significantly higher in deaths of AECOPD compared to survivors. Correspondingly, the mortality rates were significantly increased in patients with high PCT level. However, according to the multivariate analysis of the included studies, PCT levels were not a useful predictive indicator of mortality in AECOPD. Subgroup analysis showed PCT levels are associated with an increased risk of mortality in patients with mild to moderate AECOPD.

There are several reasons that may explain these differences. First, It may be that because of PCT are predominantly increased in exacerbations caused by bacterial infections which is not the only factor leading to the exacerbation of COPD. In fact, only approximately 50% of exacerbations of COPD are associated with the isolation of bacteria from the lower respiratory tract [28, 34, 35]. Second, the analysis showed high statistical heterogeneity between studies ($I^2 = 79\%$), which may caused by differences in confounding factors, statistical analysis, determination of PCT and the definition of COPD. And the weights of included studies range from 1.0% to 53.3%. PCT also showed an association with the risk of death in severe AECOPD, when excluding study with the lowest weight. In addition, an individual patient data meta-analysis done by Kutz et al. showed that PCT level was a good indicator of mortality in patients with AECOPD, with an adjusted odds ratio of 6.12(95% CI: 2.46–15.18; $p < 0.001$) [18]. In the future, the use of PCT should be further assessed in large, well-designed, prospective studies.

This study was subject to several limitations First, all included trials were conducted in Europe and Asia, thereby limiting the generalizability of the findings. Second, the multivariate analysis employed in various studies revealed substantial disparities in the range of adjusting factors incorporated, which may hinder the direct comparison of results. Finally, although our meta-analysis was solely reliant on published studies, it is acknowledged that publication bias remains a significant challenge, and potentially introduce a certain degree of bias in the overall analysis.

Conclusions

Our meta-analysis shows that procalcitonin levels at the time of hospital admission for mild to moderate AECOPD are positively correlated with mortality.

Supporting information

S1 Checklist. PRISMA 2020 checklist.
(DOCX)

S1 File. MOOSE checklist.
(PDF)

S1 Table. Search strategy.
(DOC)

S2 Table. Studies identified after excluding duplications.
(DOCX)

S3 Table. Raw data used in current meta-analysis.
(DOCX)

S4 Table. Risk of bias for each study.
(DOC)

Author Contributions

Conceptualization: Pang Qiyuan, Zhou Xun.

Data curation: Lin Changyang, Li Gaigai, Qiu Ju.

Formal analysis: Pang Qiyuan, Lin Changyang.

Methodology: Lin Changyang, Li Gaigai, Qiu Ju.

Software: Qiu Ju.

Writing – original draft: Pang Qiyuan.

Writing – review & editing: Zhou Xun.

References

1. Halpin DMG, Criner GJ, Papi A, Singh D, Anzueto A, Martinez FJ, et al. Global Initiative for the Diagnosis, Management, and Prevention of Chronic Obstructive Lung Disease. The 2020 GOLD Science Committee Report on COVID-19 and Chronic Obstructive Pulmonary Disease. *Am J Respir Crit Care Med*. 2021; 203(1):24–36. <https://doi.org/10.1164/rccm.202009-3533SO> PMID: 33146552
2. Singh D, Agusti A, Anzueto A, Barnes PJ, Bourbeau J, Celli BR, et al. Global Strategy for the Diagnosis, Management, and Prevention of Chronic Obstructive Lung Disease: the GOLD science committee report 2019. *The European respiratory journal*. 2019; 53(5). <https://doi.org/10.1183/13993003.00164-2019> PMID: 30846476
3. Rana JS, Khan SS, Lloyd-Jones DM, Sidney S. Changes in Mortality in Top 10 Causes of Death from 2011 to 2018. *J Gen Intern Med*. 2021; 36(8):2517–8. <https://doi.org/10.1007/s11606-020-06070-z> PMID: 32705476
4. Rossaki FM, Hurst JR, van Gemert F, Kirenga BJ, Williams S, Khoo EM, et al. Strategies for the prevention, diagnosis and treatment of COPD in low- and middle- income countries: the importance of primary care. *Expert Rev Respir Med*. 2021; 15(12):1563–77. <https://doi.org/10.1080/17476348.2021.1985762> PMID: 34595990
5. Hurst JR, Skolnik N, Hansen GJ, Anzueto A, Donaldson GC, Dransfield MT, et al. Understanding the impact of chronic obstructive pulmonary disease exacerbations on patient health and quality of life. *Eur J Intern Med*. 2020; 73:1–6. <https://doi.org/10.1016/j.ejim.2019.12.014> PMID: 31954592
6. Hoogendoorn M, Hoogenveen RT, Rutten-van Molken MP, Vestbo J, Feenstra TL. Case fatality of COPD exacerbations: a meta-analysis and statistical modelling approach. *The European respiratory journal*. 2011; 37(3):508–15. <https://doi.org/10.1183/09031936.00043710> PMID: 20595157

7. Singanayagam A, Schembri S, Chalmers JD. Predictors of mortality in hospitalized adults with acute exacerbation of chronic obstructive pulmonary disease. *Annals of the American Thoracic Society*. 2013; 10(2):81–9. <https://doi.org/10.1513/AnnalsATS.201208-043OC> PMID: 23607835
8. Guo Y, Zhang T, Wang Z, Yu F, Xu Q, Guo W, et al. Body mass index and mortality in chronic obstructive pulmonary disease: A dose-response meta-analysis. *Medicine*. 2016; 95(28):e4225. <https://doi.org/10.1097/MD.0000000000004225> PMID: 27428228
9. Maruna P, Nedelnikova K, Gurlich R. Physiology and genetics of procalcitonin. *Physiol Res*. 2000; 49 Suppl 1:S57–61. PMID: 10984072
10. Hall SF, Wright NC, Wolinsky FD, Lou Y, Edmonds S, Roblin D, et al. The prevalence of overtreatment of osteoporosis: results from the PAADRN trial. *Archives of Osteoporosis*. 2018; 13(1). <https://doi.org/10.1007/s11657-018-0517-6> PMID: 30267162
11. Liu D, Su L, Han G, Yan P, Xie L. Prognostic Value of Procalcitonin in Adult Patients with Sepsis: A Systematic Review and Meta-Analysis. *PloS one*. 2015; 10(6):e0129450. <https://doi.org/10.1371/journal.pone.0129450> PMID: 26076027
12. Liu D, Su LX, Guan W, Xiao K, Xie LX. Prognostic value of procalcitonin in pneumonia: A systematic review and meta-analysis. *Respirology*. 2016; 21(2):280–8. <https://doi.org/10.1111/resp.12704> PMID: 26662169
13. Liberati A, Altman DG, Tetzlaff J, Mulrow C, Gotzsche PC, Ioannidis JP, et al. The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate healthcare interventions: explanation and elaboration. *BMJ*. 2009; 339:b2700. <https://doi.org/10.1136/bmj.b2700> PMID: 19622552
14. Stroup DF, Berlin JA, Morton SC, Olkin I, Williamson GD, Rennie D, et al. Meta-analysis of observational studies in epidemiology: a proposal for reporting. Meta-analysis Of Observational Studies in Epidemiology (MOOSE) group. *JAMA*. 2000; 283(15):2008–12. <https://doi.org/10.1001/jama.283.15.2008> PMID: 10789670
15. Stolz D, Christ-Crain M, Morgenthaler NG, Miedinger D, Leuppi J, Müller C, et al. Plasma pro-adrenomedullin but not plasma pro-endothelin predicts survival in exacerbations of COPD. *Chest*. 2008; 134(2):263–72. <https://doi.org/10.1378/chest.08-0047> PMID: 18490402
16. Rammaert B, Verdier N, Cavestri B, Nseir S. Procalcitonin as a prognostic factor in severe acute exacerbation of chronic obstructive pulmonary disease. *Respirology*. 2009; 14(7):969–74. <https://doi.org/10.1111/j.1440-1843.2009.01597.x> PMID: 19659517
17. Zuur-Telgen MC, Brusse-Keizer MGJ, VanderValk PDLPM, Van Der Palen J, Kerstjens Dr HAM, Hendrix Dr MGR. Stable-state midrange-proadrenomedullin level is a strong predictor of mortality in patients with COPD. *Chest*. 2014; 145(3):534–41. <https://doi.org/10.1378/chest.13-1063> PMID: 27845633
18. Kutz A, Briel M, Christ-Crain M, Stolz D, Bouadma L, Wolff M, et al. Prognostic value of procalcitonin in respiratory tract infections across clinical settings. *Critical care (London, England)*. 2015; 19:74. <https://doi.org/10.1186/s13054-015-0792-1> PMID: 25887979
19. Grolmund E, Kutz A, Marlowe RJ, Vogeli A, Alan M, Christ-Crain M, et al. Long-term Prognosis in COPD Exacerbation: Role of Biomarkers, Clinical Variables and Exacerbation Type. *Copd*. 2015; 12(3):295–305. <https://doi.org/10.3109/15412555.2014.949002> PMID: 25230352
20. Ergan B, Sahin AA, Topeli A. Serum Procalcitonin as a Biomarker for the Prediction of Bacterial Exacerbation and Mortality in Severe COPD Exacerbations Requiring Mechanical Ventilation. *Respiration*. 2016; 91(4):316–24. <https://doi.org/10.1159/000445440> PMID: 27081845
21. Flattet Y, Garin N, Serratrice J, Perrier A, Stirnemann J, Carballo S. Determining prognosis in acute exacerbation of COPD. *International Journal of COPD*. 2017; 12:467–75. <https://doi.org/10.2147/COPD.S122382> PMID: 28203070
22. Yao C, Wang L, Shi F, Chen R, Li B, Liu W, et al. Optimized combination of circulating biomarkers as predictors of prognosis in AECOPD patients complicated with heart failure. *International Journal of Medical Sciences*. 2021; 18(7):1592–9. <https://doi.org/10.7150/ijms.52405> PMID: 33746575
23. Stolz D, Christ-Crain M, Morgenthaler NG, Leuppi J, Miedinger D, Bingisser R, et al. Copeptin, C-reactive protein, and procalcitonin as prognostic biomarkers in acute exacerbation of COPD. *Chest*. 2007; 131(4):1058–67. <https://doi.org/10.1378/chest.06-2336> PMID: 17426210
24. Ceylan O, Ucgun I, Us T, Kasifoglu N, Kiremitci A, Demircan F, et al. The role of serum procalcitonin and CRP levels in determining of etiology and outcome in acute exacerbations of COPD. *Intensive Care Medicine Experimental*. 2015; 3. <https://doi.org/10.1186/2197-425X-3-S1-A796>
25. Gong C, Yang Y, Chen M, Xie Z. Effect of procalcitonin on the prognosis of patients with COPD. *Bio-medical reports*. 2020; 12(6):313-8. <https://doi.org/10.3892/br.2020.1298> PMID: 32382415
26. Yu X, Zhu GP, Cai TF, Zheng JY. Establishment of risk prediction model and risk score for in-hospital mortality in patients with AECOPD. *Clinical Respiratory Journal*. 2020; 14(11):1090–8. <https://doi.org/10.1111/crj.13246> PMID: 32757441

27. Higgins JP, Thompson SG. Quantifying heterogeneity in a meta-analysis. *Stat Med*. 2002; 21(11):1539–58. <https://doi.org/10.1002/sim.1186> PMID: 12111919
28. Wedzicha JA, Seemungal TA. COPD exacerbations: defining their cause and prevention. *Lancet*. 2007; 370(9589):786–96. [https://doi.org/10.1016/S0140-6736\(07\)61382-8](https://doi.org/10.1016/S0140-6736(07)61382-8) PMID: 17765528
29. Dixit D, Bridgeman MB, Andrews LB, Narayanan N, Radbel J, Parikh A, et al. Acute exacerbations of chronic obstructive pulmonary disease: diagnosis, management, and prevention in critically ill patients. *Pharmacotherapy*. 2015; 35(6):631–48. <https://doi.org/10.1002/phar.1599> PMID: 26032691
30. Kibe S, Adams K, Barlow G. Diagnostic and prognostic biomarkers of sepsis in critical care. *J Antimicrob Chemother*. 2011; 66 Suppl 2:ii33–40. <https://doi.org/10.1093/jac/dkq523> PMID: 21398306
31. Lacoma A, Prat C, Andreo F, Lores L, Ruiz-Manzano J, Ausina V, et al. Value of procalcitonin, C-reactive protein, and neopterin in exacerbations of chronic obstructive pulmonary disease. *Int J Chron Obstruct Pulmon Dis*. 2011; 6:157–69. <https://doi.org/10.2147/COPD.S16070> PMID: 21468168
32. Pazarli AC, Koseoglu HI, Doruk S, Sahin S, Etikan I, Celikel S, et al. Procalcitonin: Is it a predictor of noninvasive positive pressure ventilation necessity in acute chronic obstructive pulmonary disease exacerbation? *Journal of Research in Medical Sciences*. 2012; 17(11):1047–51. PMID: 23833579
33. Wang J, Shang H, Yang X, Guo S, Cui Z. Procalcitonin, C-reactive protein, PaCO₂, and noninvasive mechanical ventilation failure in chronic obstructive pulmonary disease exacerbation. *Medicine (Baltimore)*. 2019; 98(17):e15171. <https://doi.org/10.1097/MD.00000000000015171> PMID: 31027061
34. Sethi S. Infectious etiology of acute exacerbations of chronic bronchitis. *Chest*. 2000; 117(5 Suppl 2):380S–5S. https://doi.org/10.1378/chest.117.5_suppl_2.380s PMID: 10843981
35. Beasley V, Joshi PV, Singanayagam A, Molyneaux PL, Johnston SL, Mallia P. Lung microbiology and exacerbations in COPD. *Int J Chron Obstruct Pulmon Dis*. 2012; 7:555–69. <https://doi.org/10.2147/COPD.S28286> PMID: 22969296